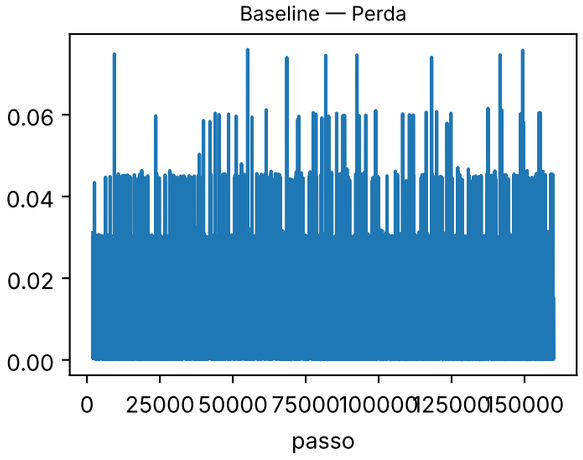
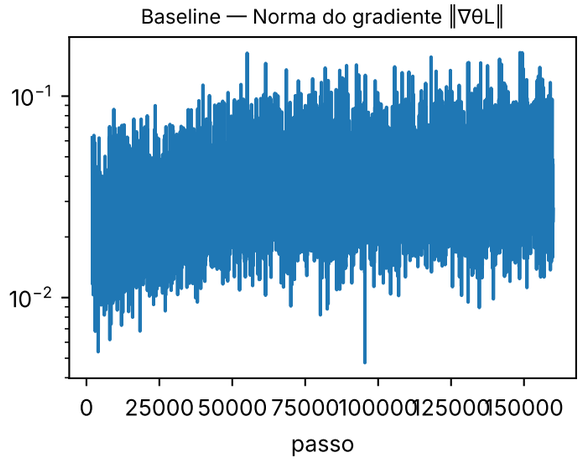
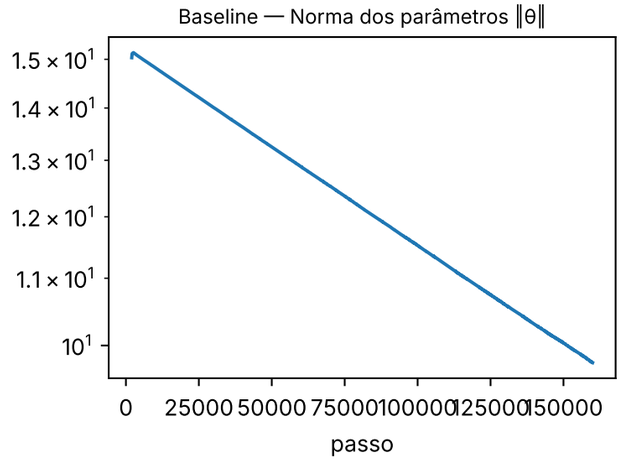
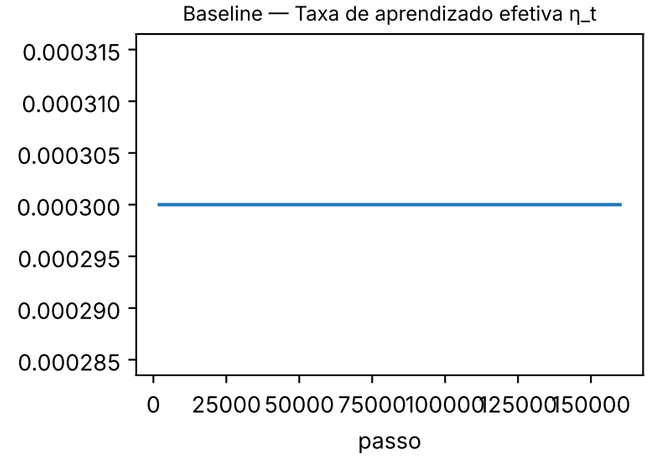
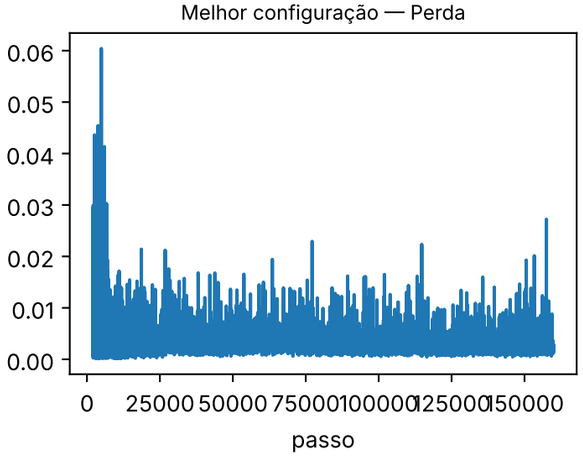
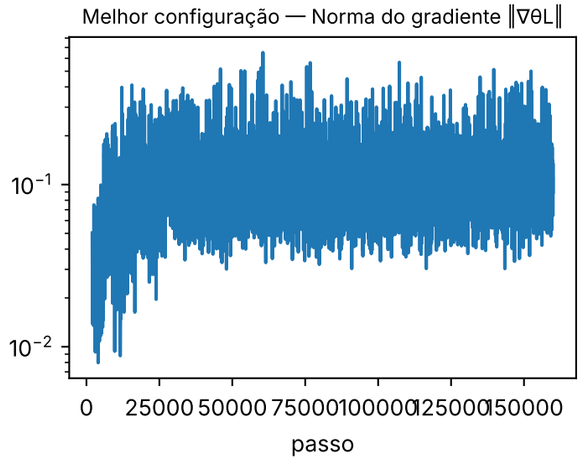

Baseline — Perda
0.06
0.04
0.02
0.00
0
25000
50000
75000
100000
125000
150000
passo

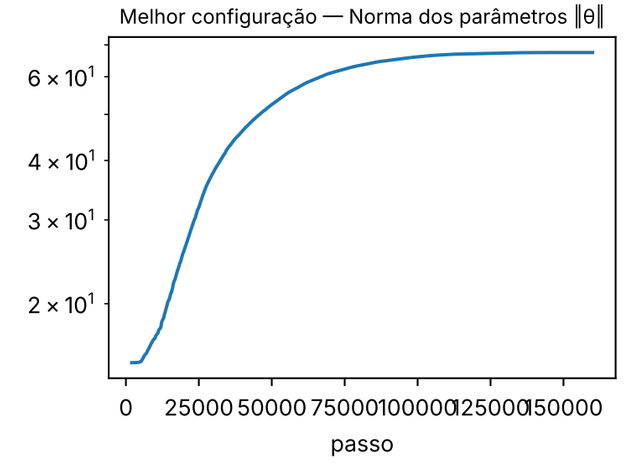

Baseline — Norma do gradiente ‖∇θL‖
10^{-1}
10^{-2}
0
25000
50000
75000
100000
125000
150000
passo

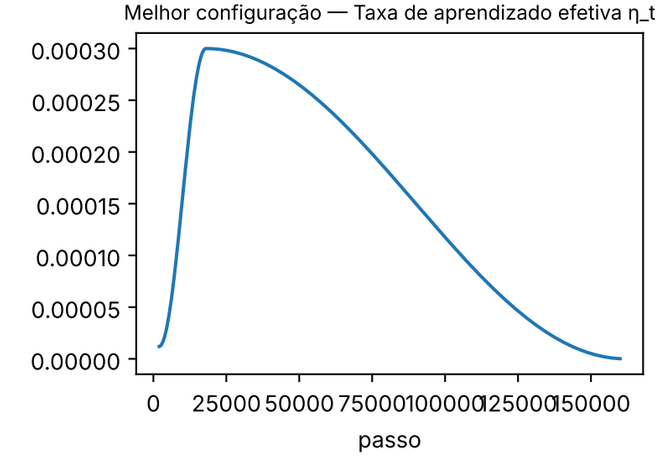

Baseline — Norma dos parâmetros ‖θ‖
1.5×10^1
1.4×10^1
1.3×10^1
1.2×10^1
1.1×10^1
10^1
0
25000
50000
75000
100000
125000
150000
passo

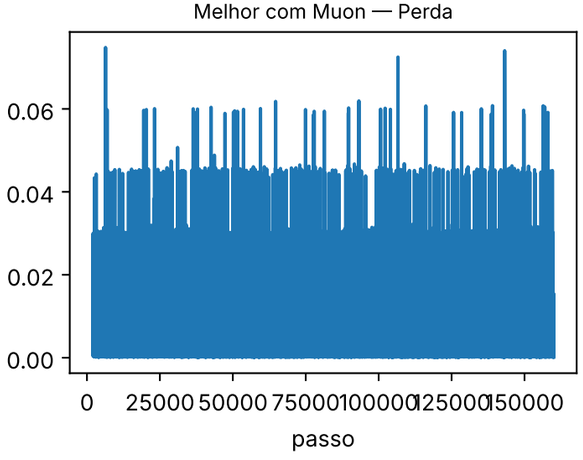

Baseline — Taxa de aprendizado efetiva η_t
0.000315
0.000310
0.000305
0.000300
0.000295
0.000290
0.000285
0
25000
50000
75000
100000
125000
150000
passo

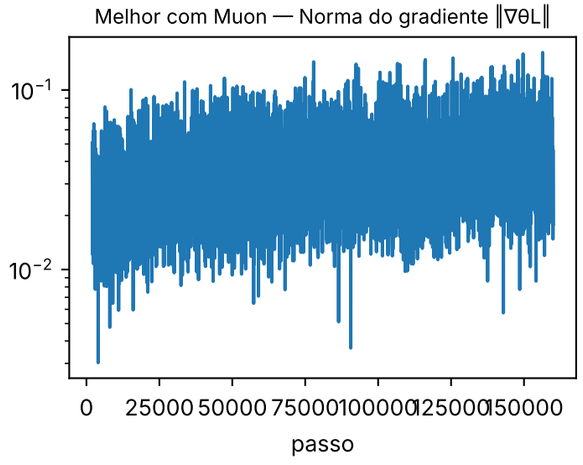

Melhor configuração — Perda
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
25000
50000
75000
100000
125000
150000
passo

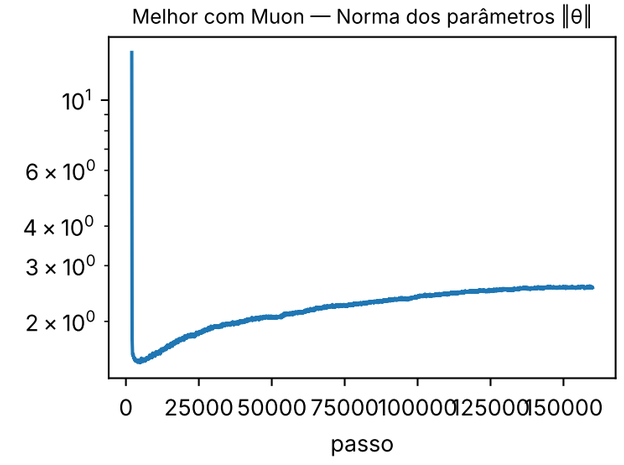

Melhor configuração — Norma do gradiente ‖∇θL‖
10^{-1}
10^{-2}
0
25000
50000
75000
100000
125000
150000
passo

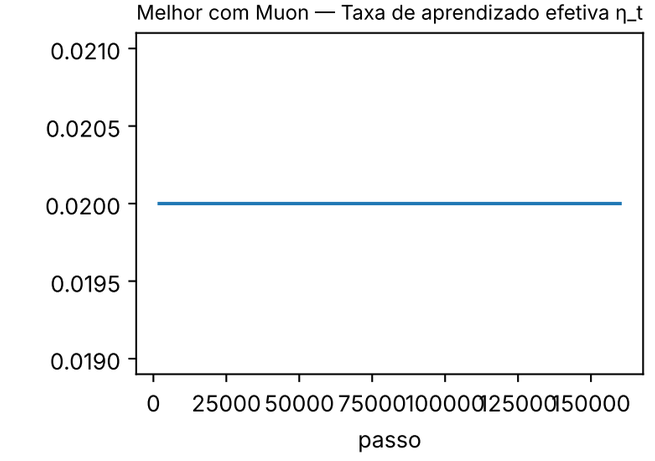

Melhor configuração — Norma dos parâmetros ‖θ‖
6×10^1
4×10^1
3×10^1
2×10^1
0
25000
50000
75000
100000
125000
150000
passo

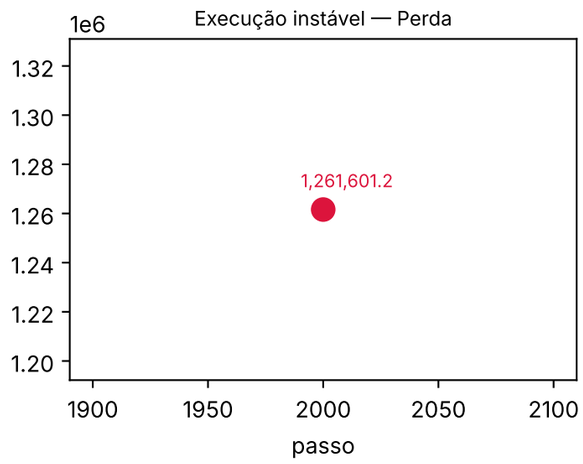

Melhor configuração — Taxa de aprendizado efetiva η_t
0.00030
0.00025
0.00020
0.00015
0.00010
0.00005
0.00000
0
25000
50000
75000
100000
125000
150000
passo

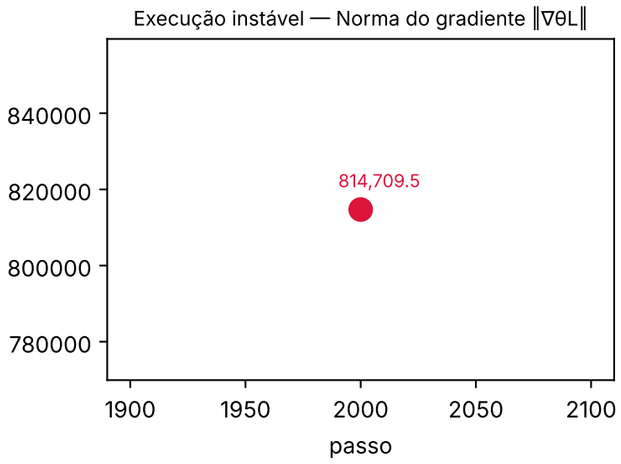

Melhor com Muon — Perda
0.06
0.04
0.02
0.00
0
25000
50000
75000
100000
125000
150000
passo

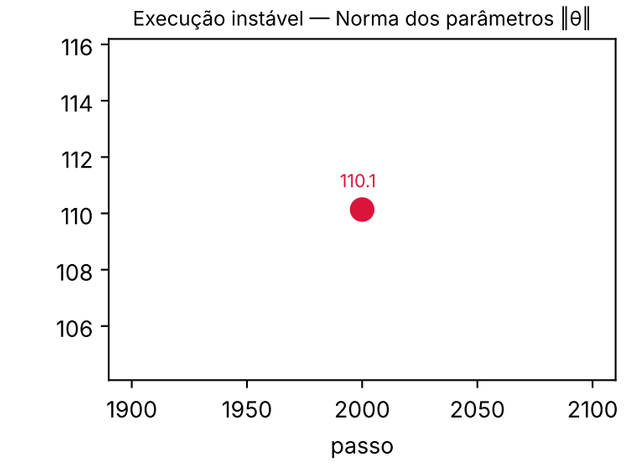

Melhor com Muon — Norma do gradiente ‖∇θL‖
10^{-1}
10^{-2}
0
25000
50000
75000
100000
125000
150000
passo

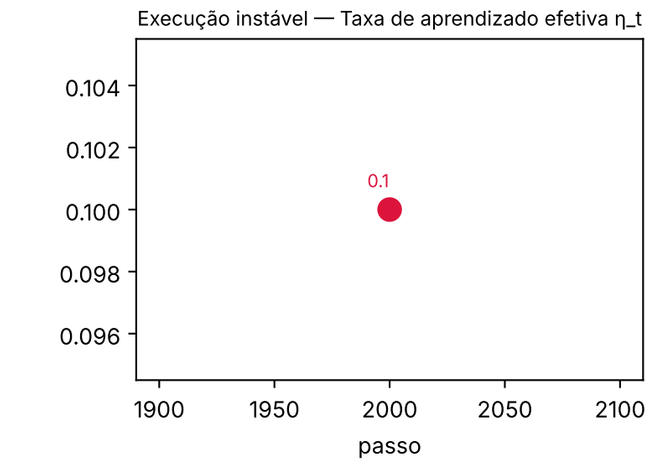

Melhor com Muon — Norma dos parâmetros ‖θ‖
10^1
6×10^0
4×10^0
3×10^0
2×10^0
0
25000
50000
75000
100000
125000
150000
passo

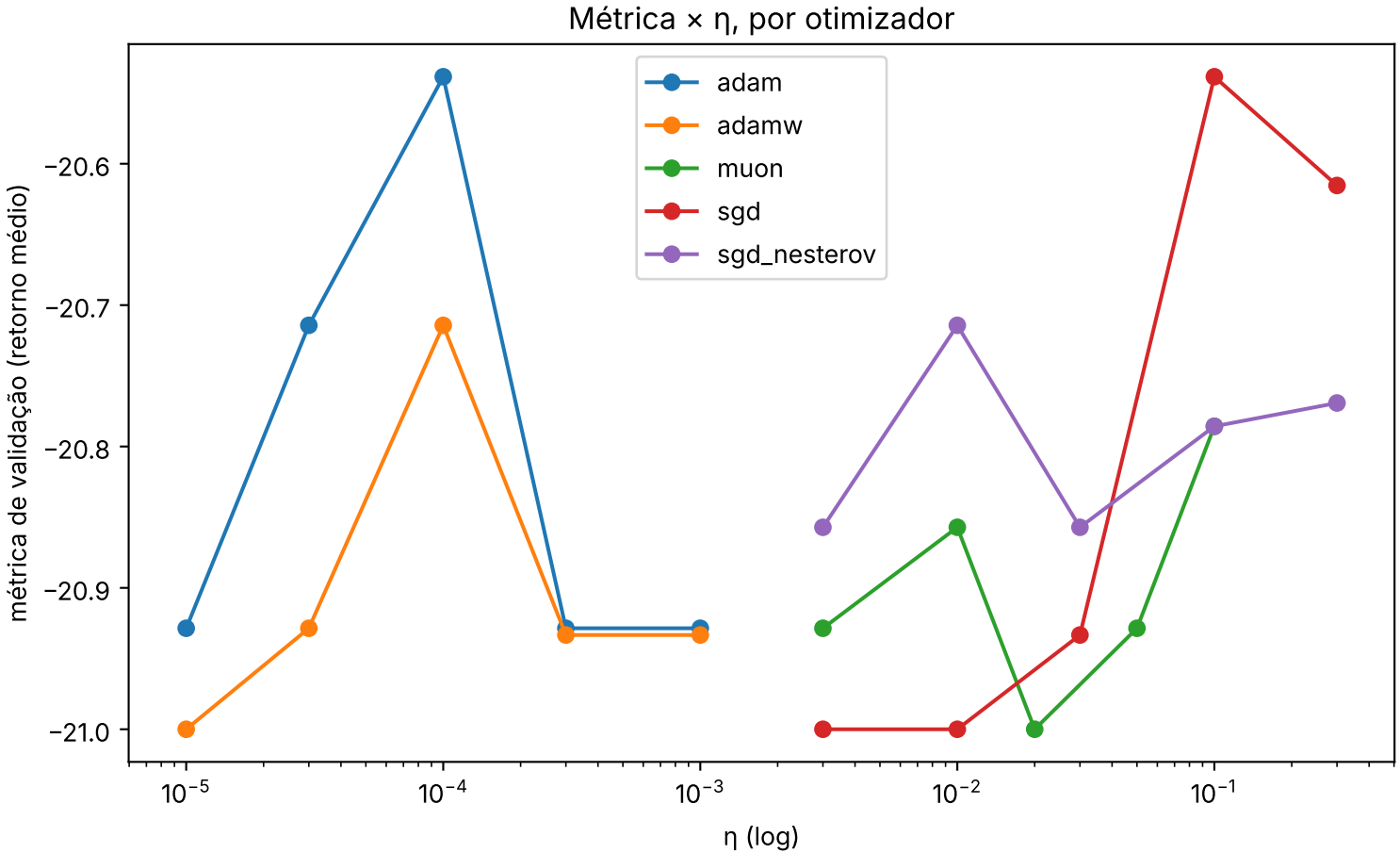

Melhor com Muon — Taxa de aprendizado efetiva η_t
0.0210
0.0205
0.0200
0.0195
0.0190
0
25000
50000
75000
100000
125000
150000
passo


Execução instável — Perda
1e6
1.32
1.30
1.28
1.26
1.24
1.22
1.20
1,261,601.2
1900
1950
2000
2050
2100
passo


Execução instável — Norma do gradiente ‖∇θL‖
840000
820000
800000
780000
814,709.5
1900
1950
2000
2050
2100
passo


Execução instável — Norma dos parâmetros ‖θ‖
116
114
112
110
108
106
110.1
1900
1950
2000
2050
2100
passo


Execução instável — Taxa de aprendizado efetiva η_t
0.104
0.102
0.100
0.098
0.096
0.1
1900
1950
2000
2050
2100
passo


Métrica × η, por otimizador
adam
adamw
muon
sgd
sgd_nesterov
métrica de validação (retorno médio)
−20.6
−20.7
−20.8
−20.9
−21.0
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
η (log)